

# Ardeshir Aidun

Los Gatos, CA | addieaidun@gmail.com | (408) 656-9422 | linkedin.com/in/ardeshir-aidun | github.com/Ardeshir-Aidun

## PROFESSIONAL SUMMARY

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Electrical Engineering student at UC Santa Barbara seeking internship and research opportunities in embedded systems, hardware design, and electrical engineering. Experienced in microcontroller programming, circuit prototyping, bench testing, and data analysis, with a track record of designing and debugging real hardware systems end to end.

## EDUCATION

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### University of California, Santa Barbara

Bachelor of Science, Electrical Engineering. GPA: 3.72.

Expected June 2028

Relevant Coursework: Digital Design, Analog and Digital Circuits Fundamentals, Signal Analysis, Robotics: Control, Quantum Photonics, Research Process for Engineers

## EXPERIENCE

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### Member

Oct. 2024 – Present

IEEE - UCSB Chapter

Goleta, CA

- Applied soldering techniques and PCB design concepts through hands-on technical workshops
- Assembled and troubleshot circuits, building practical experience in component assembly and prototyping

### Guest Advocate

June 2025 – June 2026

Target

San Jose, CA

- Coordinated order pickups and curbside deliveries, resolving fulfillment and self-checkout issues as they arose

### Leader

Aug. 2022 – May 2024

Link Crew

Los Gatos, CA

- Guided incoming students through orientation and campus acclimation activities
- Monitored student progress monthly and provided personal follow-up for those facing academic or social challenges

## PROJECTS

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### ESP32 IoT Mesh Network: Wireless Telemetry and Rollback-Safe OTA Updates

June 2026 – July 2026

- Designed and built a 3-node wireless mesh network using ESP32 microcontrollers, with sensor nodes reporting live temperature and humidity data to a central gateway over ESP-NOW
- Architected a wireless OTA firmware update system using WiFi and HTTP, leveraging the ESP32's dual-partition bootloader to safely stage and verify updates without disrupting the running device
- Validated fault-tolerant recovery by deliberately deploying broken firmware and confirming automatic rollback to the last known-good version with zero manual intervention

### Ultrasonic Sensor Accuracy and Noise Characterization

May 2026 – June 2026

- Designed and executed a bench test protocol to characterize accuracy, noise, and drift of an HC-SR04 sensor across a 5-30 cm range using an Arduino Uno
- Built a Python data pipeline to log serial sensor data and compute statistical metrics (mean error, standard deviation, drift rate), visualized with matplotlib
- Identified a distance-dependent measurement bias and noise discrepancy between test protocols, documenting findings in a written report

### Automatic Night Light

April 2026 – April 2026

- Designed and built an automated nightlight prototype using an Arduino, a photoresistor (LDR), and an LED circuit
- Programmed real-time ambient light monitoring in C++ using analog input to dynamically control LED output states
- Debugged hardware connections and serial communication errors using the Serial Monitor to optimize system performance

### Bike Indicator Lights

Oct. 2024 – Dec. 2024

- Designed and built a custom microcontroller system using an Arduino Uno and push buttons to control left, right, and hazard light functions on an LED strip
- Programmed custom control logic for directional signaling and emergency flashers to improve rider visibility

## SKILLS

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**Hardware:** Microcontrollers, Circuit Design, PCB Design, Soldering, Prototyping, Bench Testing, Oscilloscope, LTspice

**Software and Programming:** C++, Python, MATLAB, Simulink, Git/GitHub

**Communication Protocols:** UART, ESP-NOW, WiFi/HTTP

**Technical Concepts:** Digital Design, Signal Analysis, Boolean Algebra